

very important role. Naresh Narasimhan, Pacific technical marketing manager, Tektronix, says, "The highest-bandwidth oscilloscope is of little use to the user if the signal cannot be tapped and directed to the device."

Connectivity to PC

Computer has become an inseparable part of test and measurement instrumentation. Digitisation has made it easier to take the oscilloscope signal on a PC. Bruce Tulloch, director-software development, BitScope, believes, "Products that leverage power of personal computers, netbooks and tablets are expected to become even more common. These are generally more useful than traditional standalone oscilloscopes because they can be produced at lower cost, are more flexible in use, and can meet or exceed the specifications of many standalone solutions."

"In earlier days, analogue oscilloscope required single-shot external signal out and a special camera was required to capture the image. With the advent of digitisation, the same signal is captured in the memory and can be seen on the digital storage oscilloscope (DSO) screen or taken to a PC via USB, RS-232, GPIB, LAN, etc," shares Neelam Kumar, executive director, Aplab.

"The scopes come with USB host feature, so the waveform image can be captured directly onto a USB flash drive without the need of PC," adds Kumar.

"Analogue scopes have also entered the area of Ethernet connectivity and data-link applications," informs J.K. Baldua, director-technical, Scientech Technologies.

High-speed interface for data transfer

Oscilloscopes help capture signal waveforms and analyse them for performance. The increasing complexity



Scientech's Caddo 851 mixed-signal oscilloscope

of the signals and the challenges associated with signal integrity and high-speed data acquisition have stirred up the performance requirements of digital scopes.

Bhatia shares, "The oscilloscope market is now witnessing increasing adoption of high-speed serial data bus technologies like PCI Express, serial advanced technology attachment (SATA) and HDMI. As the memory depth of digital storage oscilloscope has increased to 2 giga-points, these interfaces facilitate faster data transfers."

GPIB phases out, USB takes in

USB has almost replaced GPIB in instrumentation control. The plug-and-play capability of USB is widely used in many form factors in oscilloscopes. Today, users demand performance along with ease of operation. As USB is the most commonly adopted standard in the PC world, you do not have to write the code or install drivers separately while connecting a hardware using USB port to your computer.

USB PC-based oscilloscopes eliminate the need for an external power supply and can be conveniently used with a laptop or notebook computer. Standalone box oscilloscopes offer more than one USB port for connectivity to other equipment as well as computer.

Bhatia shares, "In some of the older aerospace and defence applications, when the automated test algorithms for controlling the overall system use

GPIB for control, you may require a USB-to-GPIB converter if you use a latest oscilloscope with USB port." Many companies including Agilent and National Instruments provide USB-to-GPIB and GPIB-to-high-speed-USB connectors.

Although USB offers all the flexibility and ease of operation, there are some issues around it. Nandini Subramanya, marketing

communications manager, National Instruments, informs, "USB 2.0 does not necessarily mean high speed as USB standards provide different data rates. When a device is called USB 2.0-compatible, it supports the low- (1.5Mbps) and full-speed (12Mbps) transfer rates. To ensure greater transfer rates, you should purchase an instrument with high-speed USB (480Mbps).

"There are two types of USB ports. Type A ports are flat and elongated. Normally found on computers, these are typically used to save data to USB memory sticks and not for instrument control. For controlling your instrument through a PC, you need Type B device port, which is more like a square shape."

Connecting scope in network

When it comes to automated test applications, Subramanya cites, "Oscilloscopes are just one of the many instruments required to enable the multiple types of measurements needed to verify the functionality of today's units under test." Engineers demand high-performance, modular, flexible systems, with easy upgradeability to the latest commercial-off-the-shelf (COTS) technology and reusability along with the option to control remotely.

LAN eXtensions for Instrumentation (LXI) based instruments come in compact, flexible packages with high-speed capabilities and reliable measurements. Bhatia shares, "Ethernet LAN enables wired as well as wireless